

MAYAR HANY RAFIK

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PROFESSIONAL SUMMARY

Entry-level Data Engineer professional with hands-on experience building production-grade ETL/ELT pipelines, analytical data warehouses, and machine learning-powered dashboards. Proficient in Python, SQL, Apache Spark, and Hadoop, with proven ability to orchestrate end-to-end data workflows using Dagster and deploy interactive applications via Streamlit and Plotly Dash. Track record of processing 50,000+ records across supply chain, HR, medical, and NASA climate datasets. Adept at translating raw data into actionable business insights through Power BI, Tableau, and custom dashboards. Currently pursuing a degree in Business Intelligence & Analytical Systems (GPA 3.82/4.0).

TECHNICAL SKILLS

Programming: Python, SQL, R, Bash

Big Data: Apache Spark, Hadoop, Hive, HBase, Elasticsearch

Databases: Snowflake, PostgreSQL, SQL Server, MySQL

Data Processing: Pandas, NumPy, Scikit-learn, ETL/ELT Development, Feature Engineering, Statistical Analysis, SMOTE

Orchestration: Dagster, Apache Spark Pipelines

Cloud & APIs: Huawei Cloud (HCIA Certified), REST API Integration

Visualization: Power BI, Tableau, Plotly, Dash, Streamlit

ML / AI: RandomForest, Logistic Regression, SVM, K-Means, Deep Learning (NVIDIA Certified), NLP / TF-IDF

Tools: Git, Jupyter, Google Colab, VS Code, Docker (basic), Report Builder

PROFESSIONAL EXPERIENCE

AI & Machine Learning Trainee | National Telecommunication Institute (NTI)

Cairo, Egypt

Jul 2025 – Aug 2025

- Built ML feature engineering pipelines transforming raw HR and customer data into model-ready feature sets, including categorical encoding, numerical scaling, and SMOTE for class imbalance.
- Developed automated prediction pipelines and deployed real-time monitoring dashboards using Streamlit and Plotly for model serving and performance tracking.
- Achieved 85%+ prediction accuracy on bonus eligibility classification using RandomForestClassifier.

BI Developer Intern | Housing and Development Bank (HDB)

Cairo, Egypt

Jul 2025

- Developed automated reporting solutions using SQL Server Report Builder, consolidating data from multiple banking systems into structured management reports.
- Authored complex SQL queries with JOINS, aggregations, and window functions for financial data extraction, significantly reducing manual report preparation effort.
- Explored bank IT infrastructure including data center components, network architecture, cybersecurity protocols, and threat detection systems.

Data Analytics Trainee | Digital Egypt Pioneers Initiative (DEPI)

Cairo, Egypt

Oct 2024 – May 2025

- Designed Python ETL scripts extracting data from CSV, JSON, and API sources; transformed with Pandas and loaded into analytical databases.
- Led a team project analyzing supply chain datasets (logistics, customer behavior, profitability) across 50,000+ records.
- Implemented automated data cleaning and quality checks to ensure accurate loading into downstream analytics systems.

Big Data Analytics Trainee | National Telecommunication Institute (NTI)

Cairo, Egypt

Aug 2024 – Sep 2024

- Processed large-scale datasets using Apache Spark and Hive on a Hadoop distributed cluster environment.
- Implemented data transformation pipelines using HBase for NoSQL storage and Elasticsearch for search indexing.
- Developed batch SQL queries on Hive for aggregation across multi-terabyte datasets.
- Earned NTI & Huawei HCIA-level Big Data certifications upon completion.

Power BI & Tableau Trainee | CognoRise

Remote (India)

Oct 2024 – Nov 2024

- Developed interactive dashboards and advanced data visualizations; strengthened DAX functions and real-time reporting proficiency.

Machine Learning Trainee | Prodigy InfoTech

Remote (India)

Oct 2024 – Nov 2024

- Applied supervised and unsupervised ML algorithms to real-world projects, enhancing feature engineering and model evaluation skills.

KEY PROJECTS

GSoC 2026 — ML4SCI DeepLense | Gravitational Lens & Dark Matter Classification

(Python, PyTorch, ResNet18, EfficientNet-B0, Scikit-learn)

- Selected for Google Summer of Code 2026 (ML4SCI organisation) to build deep learning pipelines for gravitational lens detection and dark matter substructure classification in simulated astronomical data (HSC/HST telescopes).
- Achieved AUC = 0.9909 (EfficientNet-B0) and AUC = 0.9905 (ResNet18) on a severely imbalanced binary lens-detection dataset (1:100 test ratio) by implementing PyTorch WeightedRandomSampler, weighted loss, dropout regularisation, and on-the-fly geometric augmentation.
- Delivered a 3-class dark matter substructure pipeline (no_sub / subhalo / vortex) by fine-tuning both architectures on 1-to-3-channel converted images; EfficientNet-B0 matched ResNet18 performance at ~3x fewer parameters (4.0M vs 11.2M), demonstrating parameter efficiency in a science compute environment.

Airbnb Analytics — End-to-End Price Prediction Pipeline | Orange Egypt Hackathon

(Python, Playwright, SQL Server, Power BI, R, CatBoost, Streamlit)

- Engineered a 6-component production pipeline (scrape → clean → warehouse → analyse → visualise → predict) across 251,000+ Airbnb listings from 10 European cities, achieving \$35.79M total revenue modelling coverage with an avg. satisfaction score of 87.13%.
- Built two Playwright + Chromium scrapers (CLI and Tkinter GUI) with anti-bot stealth injection and parallel ThreadPoolExecutor workers; cleaned and imputed 30+ column dataset using KNN imputation (k=3, batched by 100 rows) and feature-engineered 6 amenity booleans and a weekend indicator.
- Designed a SQL Server star schema (5 dim + 1 fact, MD5-hashed surrogate keys) with 9 business queries (ListingScore, BestDeals/WorstDeals, FeatureImpactRadar, GeoDemandHotspots, StrategyScoring + 4 more, 849 lines total); applied ANOVA (Tukey HSD post-hoc) + MANOVA (Wilks' Lambda) in R to validate room-type price drivers across satisfaction and cleanliness dimensions; delivered a 4-page Power BI dashboard (Overview, City Comparison, Pricing Analysis, Host & Amenity Insights) and a live CatBoost Regressor Streamlit app — **Live Demo**.

NASA Space Apps Challenge — WillItRain (Weather Prediction Pipeline)

NASA Space Apps Challenge — Oct 4–5, 2025 (Python, Streamlit, Snowflake, Dagster, Plotly, NASA POWER API)

- Built under NASA Space Apps Challenge (Oct 4–5, 2025) — the world's largest hackathon — an end-to-end rain prediction pipeline consuming NASA POWER climate APIs for multiple Egyptian cities, from raw API ingestion to a deployed public app.
- Orchestrated a full ELT pipeline — API extraction, transformation, and loading into Snowflake cloud data warehouse via Dagster — delivering reliable 10-day climate forecasts with rain probability and wind/temperature trend layers for end users with no data access.
- Deployed a Streamlit + Plotly interactive web app serving real-time forecast visualisations within the hackathon timeline — **Live Demo**.

Supply Chain ETL & BI Pipeline

(Python, Pandas, PostgreSQL, Plotly Dash)

- Architected an ETL pipeline ingesting logistics, customer, and profitability data from 4 source systems into a PostgreSQL analytical warehouse with fact/dimension schema.
- Built a Plotly Dash BI dashboard over 50,000+ records enabling business users to monitor and optimize supply chain KPIs.

HR Analytics & Bonus Prediction App

(Python, Streamlit, Scikit-learn, Plotly)

- Built a feature engineering pipeline for employee bonus prediction (RandomForestClassifier + SMOTE), achieving 85%+ model accuracy.
- Deployed an interactive Streamlit dashboard with Plotly charts covering salary trends, tenure analysis, gender pay gap, and department tracking — **Live Demo**.

Sentiment Analyzer Web App

(Streamlit, Scikit-learn, NLTK)

- Trained a Logistic Regression model using TF-IDF vectorization and NLP preprocessing for real-time positive/negative sentiment classification.
- Packaged and deployed the pipeline with Pickle for seamless inference; designed a custom dark + pink UI for user engagement — **Live Demo**.

Medical Cost Analysis Dashboard

(Python, Plotly Dash)

- Built an interactive Dash dashboard analyzing medical insurance datasets to uncover demographic drivers of medical charges.
- Visualized age, BMI, and cost distributions via box plots, strip plots, heatmaps, and radar charts; implemented dark/light theme toggle.

Business Intelligence Dashboard Portfolio

(Power BI, Tableau)

- Spotify Music & Podcasts Dashboard — listener engagement, most-streamed content, audience trends.
- Customer Analysis Dashboard — demographics, purchasing behavior, and profitability insights.
- Driver Performance & Logistics Dashboard — delivery KPIs, driver efficiency, and route optimization.

EDUCATION

Business Intelligence & Analytical Systems

Helwan National University | Cairo, Egypt | Sep 2023 – Expected 2027 | GPA: 3.82 / 4.0

- **Relevant Coursework:** Database Management, Data Structures, Statistical Analysis, Machine Learning, Data Visualization, Business Analytics

CERTIFICATIONS & TRAINING

- **Getting Started with Deep Learning** — NVIDIA (Aug 2025)
- Huawei HCIA Big Data Certification — Huawei / NTI (Sep 2024)
- **EF SET English Certificate 52/100 (B2 Upper Intermediate)**

LANGUAGES

Arabic: Native / Bilingual

English: Very Good (B2 Working Proficiency)